Enrollment No. 8

GANPAT UNIVERSITY

M.Sc.IT(CS)– SEM-III EXAMINATION (CBCS)

Regular– Nov.–Dec. 2023

P73A1MTA : MALWARE TAXANOMY AND ANALYSIS

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	What is malware analysis? Discuss its primary objectives and its importance in cybersecurity.	06	CO1	BTL1
2	Discuss three common types of malware and provide an overview of their functionalities.	06	CO1	BTL2
3	Explain the concept of sandboxing in malware analysis. How does sandboxing aid in analyzing and understanding malware behavior?	06	CO2	BTL4
4	Describe the key steps involved in setting up a dedicated virtual environment for malware analysis.	06	CO3	BTL3
5	Explain the concept of code emulation and disassembly in advanced static analysis.	06	CO4	BTL5
6	Explain the role of snapshotting in a malware analysis environment. How can the use of snapshots assist analysts in reverting changes made by malware and restoring the environment to a clean state?	06	CO3	BTL6
7	Discuss the process of feature extraction from malware samples for machine learning-based analysis.	06	CO5	BTL4

Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	Explain the difference between static and dynamic malware analysis techniques. Provide examples of tools used in each approach.	06	CO1	BTL3
2	Describe at least three types of strings commonly found in malware code	06	CO2	BTL3
3	Explain the concept of packing and obfuscation techniques used by malware creators. How do these techniques hinder malware analysis?	06	CO2	BTL4
Q.3	**Answer the following: (Any Two)**	**12**		
1	Outline the steps to set up a controlled and isolated network within a virtual environment to monitor malware communication without risking the integrity of the external network.	06	CO3	BTL5
2	Describe the role of Procmon (Process Monitor) in dynamic malware analysis. How does it assist analysts in monitoring system activity and tracing the behavior of potentially malicious processes?	06	CO4	BTL3
3	Elaborate on the concept of behavioral signatures in advanced static analysis. How are behavioral patterns extracted from code and utilized as indicators for identifying malware?	06	CO4	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	You are a team leader of a Cyber Security Responder Team and there is a ransomware attack on a pharma industry. You are asked to respond to the incident. Kindly Explain the process of actions by your team.	06	CO5	BTL6
2	Differentiate between polymorphic and metamorphic malware. How does advanced static analysis adapt to handle these types of malware that mutate to evade detection?	06	CO5	BTL4

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

MSc IT (Cyber Security) – III (CBCS)
Regular/Remedial – Nov.-Dec. 2023
P73A2MDS – Mobile Device Security and Forensic

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
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3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	Explain Android Manifest file in detail.	06	CO1	BTL6
2	Demonstrate the use of Burp Suite.	06	CO2	BTL4
3	Describe the Android Startup Process in detail.	06	CO2	BTL1
4	How APK Tool works? Explain all phases in detail.	06	CO3	BTL6
5	What is Passive Analysis? Explain in detail.	06	CO4	BTL2
6	Describe https proxy interception.	06	CO4	BTL1
7	Explain various types of forensic.	06	CO5	BTL3
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	Explain Android architecture in detail.	06	CO1	BTL6
2	Demonstrate Edit Text in detail.	06	CO1	BTL3
3	Describe ADB in detail.	06	CO2	BTL1
Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain insecure file storage.	06	CO3	BTL6
2	Describe content provider leakage.	06	CO3	BTL1
3	Explain Active Analysis.	06	CO4	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Describe android file system partition.	06	CO5	BTL6
2	Write a short note on AFLogical.	06	CO5	BTL5

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M.Sc.IT(Cyber Security) - III (CBCS)
Regular– Nov.-Dec. 2023
P73A3NF: NETWORK FORENSIC

Time: 3 Hours Total: 60 Marks

Instructions:

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4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	Explain the Honeypot Forensics.	06	CO1	BTL6
2	Discuss an attack graph-based frameworks.	06	CO2	BTL2
3	Discover the Nessus.	06	CO3	BTL3
4	Analyze the Snort.	06	CO4	BTL4
5	Define the NetFlow Record Format.	06	CO5	BTL1
6	Justify the NetworkMiner.	06	CO4	BTL5
7	Describe the Botnet Forensics	06	CO1	BTL1

Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	Describe the Steganography	06	CO1	BTL1
2	Discuss the generic process model for network forensics.	06	CO2	BTL2
3	Point out the challenges in network forensic analysis.	06	CO1	BTL4
Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain the Xplico in brief.	06	CO3	BTL2
2	Analyze the WebScarab.	06	CO4	BTL4
3	Identify the Nmap in detail.	06	CO4	BTL1
Q.4	**Answer the following: (Any One)**	**06**		
1	Justify the Packet Capture Format	06	CO5	BTL5

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M.Sc.IT (Cyber Security) SEM-III (CBCS)
Regular/Remedial – Nov.-Dec. 2023
P73B5ISC: Information Security Compliance Management

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
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4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	Explain the fundamental principles of Information System Security.	06	CO1	BTL1
2	Enumerate and briefly describe the key contexts covered by ISO 27001.	06	CO2	BTL2
3	Give a brief explanation of the different types of Intellectual Property Rights.	06	CO2	BTL4
4	Write a short note on HIPAA	06	CO4	BTL6
5	Explain the concept of information classification with a relevant example.	06	CO3	BTL2
6	Briefly outline the ISO 27001 certification process.	06	CO2	BTL6
7	Explain Organization of information security in brief	06	CO4	BTL5
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	What is ISMS? Discuss the benefits of ISMS in detail.	06	CO2	BTL3
2	Define Information Security and briefly explain the need for it.	06	CO3	BTL3
3	Write a short note on PDCA cycle	06	CO2	BTL2
Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain the Payment Card Industry Data Security Standard (PCI DSS) in detail.	06	CO3	BTL3
2	Provide a concise explanation about Access Control.	06	CO3	BTL6
3	Briefly discuss the various aspects of Communications and Operations Management.	06	CO4	BTL5
Q.4	**Answer the following: (Any One)**	**06**		
1	Discuss the Code of Ethics for Information Security Experts in detail.	06	CO4	BTL5
2	Define Intellectual Property Rights (IPR) and outline the advantages of IPR.	06	CO4	BTL6

**** End of the Paper ****